Calcurion Maths Worksheets

GCSE Mathematics

Topic: Pythagoras & Trigonometry

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Find the hypotenuse of a right-angled triangle with sides 3 and 4. **[2]**

2. Find the hypotenuse of a right-angled triangle with sides 5 and 12. **[2]**

3. A right-angled triangle has legs 6 and 8. Find the hypotenuse. **[2]**

4. Find the length of the side x in a right triangle with hypotenuse 10 and other side 6. **[3]**

5. Find the length of the side x in a right triangle with hypotenuse 13 and other side 5. **[3]**

6. In a right triangle, $\sin\theta = \dfrac{3}{5}$. Find $\cos\theta$. **[3]**

7. In a right triangle, $\cos\theta = \dfrac{5}{13}$. Find $\sin\theta$. **[3]**

8. Find $\tan\theta$ if $\sin\theta = \dfrac{5}{13}$ and θ is acute. **[3]**

9. Calculate $\sin 30°$. **[1]**

10. Calculate $\cos 60°$. **[1]**

11. Calculate $\tan 45°$. **[1]**

12. Find the angle θ if $\sin\theta = 0.5$ and θ is acute. **[2]**

13. Find the angle θ if $\cos\theta = \dfrac{\sqrt{3}}{2}$ and θ is acute. **[2]**

14. A ladder of length 5 m leans against a wall. The base is 3 m from the wall. How high up the wall does it reach? **[3]**

15. A right triangle has opposite side 7 and hypotenuse 25. Find $\sin\theta$. **[2]**

Section B – Hard (Q16–25)

16. Find the area of a right-angled triangle with legs 9 and 12. **[3]**

17. In triangle ABC, angle C is 90°, AC = 8, BC = 15. Find AB. **[3]**

18. Find the exact value of $\sin 45^\circ$. **[2]**

19. Find the exact value of $\cos 30^\circ$. **[2]**

20. A boat sails 12 km east then 5 km north. How far is it from the starting point? **[3]**

21. Find the angle of elevation of the top of a 20 m tower from a point 15 m from its base. **[4]**

22. In a right triangle, opposite = 12, adjacent = 5. Find the hypotenuse and $\tan\theta$. **[4]**

23. Solve $\sin\theta = 0.6$ for $0^\circ \le \theta \le 180^\circ$. **[3]**

24. A ramp rises 1.5 m over a horizontal distance of 8 m. Find the angle of the ramp to the horizontal. **[4]**

25. Prove that $\sin^2\theta + \cos^2\theta = 1$ using a right triangle of sides 3, 4, 5. **[4]**

End of Worksheet • *Show all working clearly.*